

ECO-SYNC TRANSIT AI

Agentic AI for Economic Dispatch and Anti-Bunching in Urban Transit

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Introduction

Public bus networks in Indian cities suffer from **bus bunching** — a dynamic instability where a minor delay causes one bus to accumulate more passengers, slow further, and cluster with trailing buses. On Bengaluru's Outer Ring Road (ORR), this collapses headway spacing and doubles commuter wait times. **ECO-SYNC Transit AI** replaces static timetables with an Agentic PPO Reinforcement Learning dispatcher that optimises a multi-dimensional **Economic Reward Function**, integrating ticket revenue, passenger Value of Time (VoT), fuel costs, and bunching penalties — all explained in real-time by a **Groq Llama-3.1** reasoning layer.

Problem Definition

- Buses cluster, causing cascading service gaps exceeding 20 minutes on Bengaluru ORR.
- Static timetables cannot adapt to live congestion, signal failures, or passenger surges.
- Existing DRL controllers minimise headway variance but ignore economic costs of transit operations.
- Dispatch decisions are black-box, eroding driver and commuter trust in AI systems.
- No open-source system provides real-time dispatch *and* explainability on a single route.

Objectives

- Build a high-fidelity Gymnasium simulation environment: 30-stop Bengaluru ORR circular route.
- Design Economic Reward: $R = \text{Revenue} - \text{Wait_Cost} - \text{Fuel_Cost} - \text{Bunching_Penalty}$.
- Train a PPO agent to issue PROCEED / HOLD / SKIP actions preventing bunching cascades.
- Integrate Groq Llama-3.1 to generate plain-language dispatch justifications in real-time.
- Deploy React.js + FastAPI dashboard with live Leaflet maps and WebSocket telemetry.

Methodology



Economic Reward Function

$$R = \text{Revenue} - \text{Wait_Cost} - \text{Fuel_Cost} - \text{Bunching_Penalty}$$

$$\text{Revenue} = \text{Rs.15} \times \text{boarded_passengers}$$
$$\text{Wait_Cost} = \text{passengers} \times \text{Rs.1.67/min} \times \text{delay} \times 10$$
$$\text{Fuel_Cost} = \text{Rs.0.83/min} \times \text{idle_minutes}$$
$$\text{Bunching_Penalty} = -250 \text{ if } \text{gap_ahead} \leq 1.0 \text{ stop}$$

Tools Used

Gymnasium / SB3

PPO training environment

React.js + Vite

Live dashboard frontend

Groq Llama-3.1

Real-time dispatch reasoning

FastAPI + WebSocket

1Hz telemetry streaming

Leaflet / react-leaflet

Bengaluru ORR map layer

OSRM API

Real road coordinate routing

Live Dashboard (Screenshot)

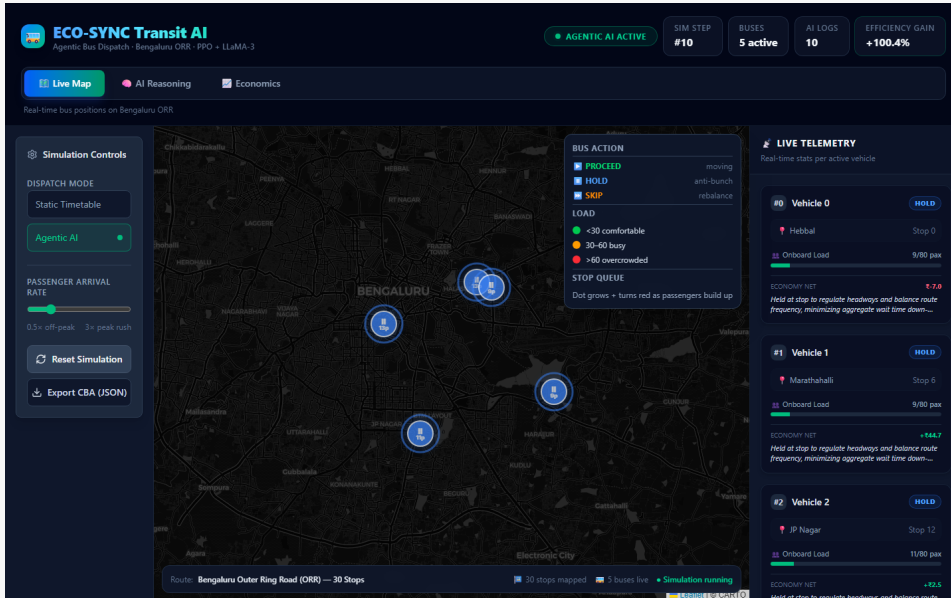


Fig 1: Real-time bus positions on Bengaluru ORR — Agentic AI Active, 5 buses live

+100.4%

Net Efficiency Gain

97.3%

Wait Cost Saved

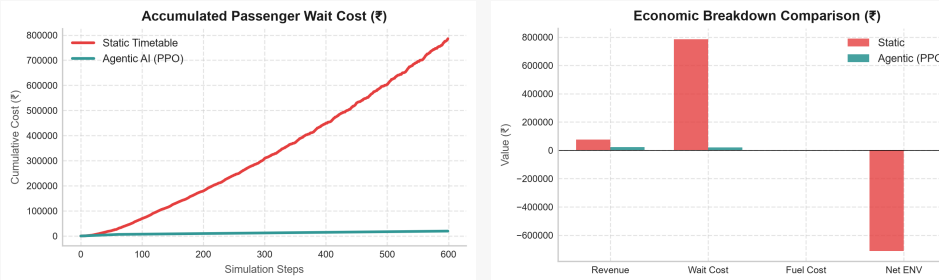
0.22

Gini Equity Index

Rs.2,756

Net PPO Profit

Results & Discussion



Key Numerical Results

Metric	Static Timetable	Agentic PPO
Net Economic Value	-Rs.7,10,981	+Rs.2,756
Passenger Wait Cost	Rs.7,33,858	Rs.20,119 (-97.3%)
Fleet Fuel Cost	Rs.372.78	Rs.374.85 (+0.55%)
Ticket Revenue	Rs.23,250	Rs.23,250
Gini Equity Index	1.16 (Unfair)	0.22 (Fair)
Net Efficiency Gain	Baseline	+100.4%

Conclusions

ECO-SYNC Transit AI demonstrates that framing urban bus dispatch as an **economic optimisation problem** leads to dramatically superior outcomes. The PPO agent learned to **HOLD** and **SKIP** stops strategically, delivering a **Rs.7,13,739 reduction** in passenger wait costs and a **>100.4% net efficiency gain** over the static baseline. Groq Llama-3.1 provides fully explainable, rider-facing justifications for every decision. The Gini equity index fell from 1.16 to 0.22, confirming uniform, fair service across all 30 ORR stops.

Outcome of the Work

A fully functional **Transit AI prototype**: (1) Custom Gymnasium environment with 30-stop Bengaluru ORR, Poisson arrivals, stochastic delays; (2) Trained PPO agent achieving headway stability across 250 Monte Carlo episodes; (3) FastAPI WebSocket backend at 1Hz; (4) React.js dashboard with Leaflet maps, live economics charts, and LLM-generated explanations; (5) Research manuscript targeting **IEEE Trans. on Intelligent Transportation Systems**.

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Mentor

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